

Project Documentation

Employee Raise Issue – Record Producer and Service Portal Integration

1. Introduction

- Project Title: Employee Raise Issue – Record Producer and Service Portal Integration
- Team Members: Individual Project (Self) – Developed, Configured, and Tested by Developer.

2. Project Overview

- Purpose: To provide employees with a centralized, user-friendly, and structured mechanism to raise workplace-related issues, replacing informal channels like emails.
- Features: Scoped Application architecture, custom table for data storage, dynamic UI policies, field dependencies, Record Producer form, dedicated Service Portal, and custom widgets.

2. IDEATION PHASE

2.1 Problem Statement Traditional workplace grievance and issue reporting relies heavily on

fragmented emails or informal verbal communication, leading to missing information, delayed tracking, and lack of accountability.

2.2 Empathy Map Canvas • Says: 'It takes too long to find who to email when my office equipment breaks.'

- Thinks: 'Are my complaints even being recorded or tracked anywhere?'
- Does: Sends multiple follow-up emails and relies on colleagues for support contacts.
- Feels: Frustrated by lack of visibility and administrative delays.

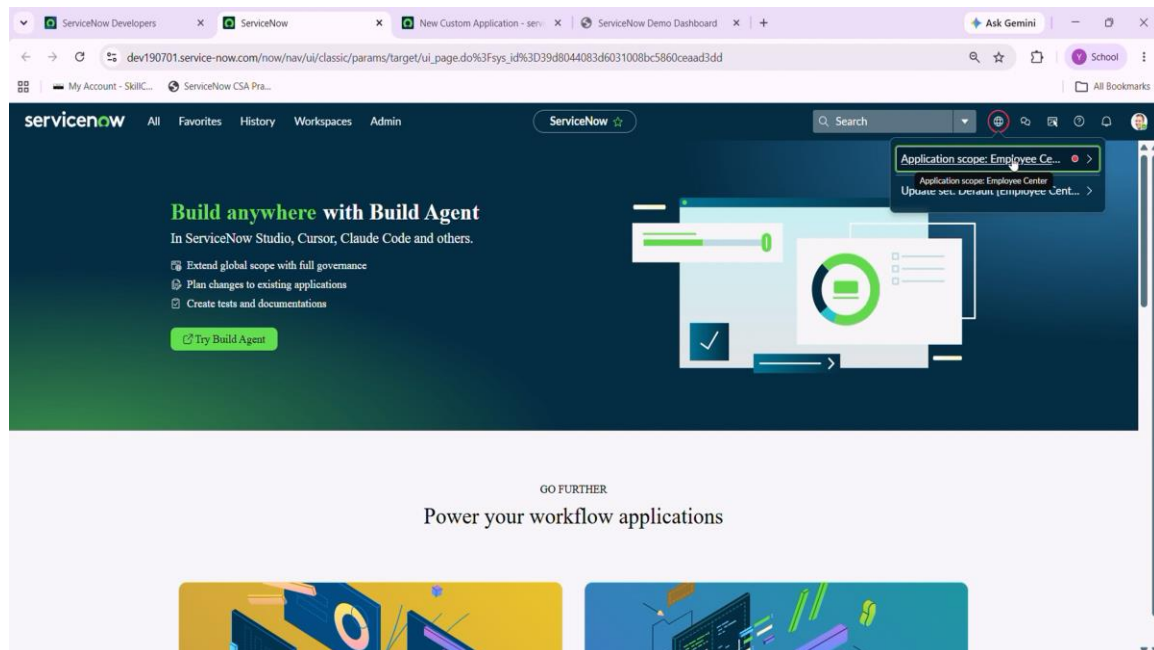
2.3 Brainstorming Explored solutions including custom email parsers, third-party ticketing tools,

and native ServiceNow Service Portal workflows. Selected the Record

Producer + Service Portal approach for optimal speed and native integration.

3. REQUIREMENT ANALYSIS • Frontend (Service Portal): Built using ServiceNow Service Portal framework, utilizing custom widgets (Left and Right widgets) and AngularJS/Bootstrap components for a responsive user interface.

- Backend (ServiceNow Core): Powered by ServiceNow server-side scripting, Business Rules, and Record Producers running within an isolated Scoped Application scoped



- Database (Custom Table): Data is structured and persisted in a dedicated custom table

The screenshot shows the 'Tables' page in the ServiceNow Developer Instance. The top navigation bar includes 'ServiceNow Developers', 'Tables | ServiceNow', 'New Custom Application - serv...', and 'ServiceNow Demo Dashboard'. The main header features the 'servicenow' logo, navigation links (All, Favorites, History, Workspaces, Admin), a search bar, and a 'Tables' button. The main content area has a light blue background with a search bar and a 'Tables Updated' button. Below this is a table with the following columns: 'Label', 'Name', 'Extends table', 'Extensible', and 'Updated'. The table lists various custom tables, including 'Employee raises issue', 'Project Task', 'Project', 'Build Agent Conversation', 'Build Agent Message', 'Event telemetry', 'Task telemetry', 'Build Agent Skill', 'Build Agent Checkpoint', 'Build Agent Skill Resource', 'Build Agent Conversation Change Log', 'Freemium Usage Log', 'IDE Extension', 'Source Artifact', 'Data Provider Cache', 'Role Combination', 'Application File to Source Artifact', 'Async Service Execution', and 'Scheduleable'. The 'Employee raises issue' table is highlighted.

Label	Name	Extends table	Extensible	Updated
Employee raises issue	x_1959081_employ_o_employee_raises_issue	(empty)	false	2026-08-10 21:24:35
Project Task	x_1959081_projec_o_project_task	(empty)	false	2026-07-28 02:10:42
Project	x_1959081_projec_o_project	(empty)	false	2026-07-28 02:07:01
Build Agent Conversation	sn_build_agent_conversation	(empty)	false	2026-07-20 03:29:20
Build Agent Message	sn_build_agent_message	(empty)	false	2026-07-20 03:29:19
Event telemetry	sn_build_agent_event_telemetry	(empty)	false	2026-07-20 03:23:45
Task telemetry	sn_build_agent_task_telemetry	(empty)	false	2026-07-20 03:23:44
Build Agent Skill	sn_build_agent_skill	(empty)	false	2026-07-20 03:23:44
Build Agent Checkpoint	sn_build_agent_checkpoint	(empty)	false	2026-07-20 03:23:43
Build Agent Skill Resource	sn_build_agent_skill_resource	(empty)	false	2026-07-20 03:23:43
Build Agent Conversation Change Log	sn_build_agent_conversation_change_log	(empty)	false	2026-07-20 03:23:42
Freemium Usage Log	sn_build_agent_freemium_usage_log	(empty)	false	2026-07-20 03:23:42
IDE Extension	sn_glider_ide_extension	(empty)	false	2026-07-20 03:23:40
Source Artifact	sn_glider_source_artifact	(empty)	false	2026-07-20 03:23:40
Data Provider Cache	sn_atf_tg_data_provider_cache	(empty)	false	2026-07-20 03:23:39
Role Combination	sn_atf_tg_role_combo	(empty)	false	2026-07-20 03:23:39
Application File to Source Artifact	sn_glider_source_artifact_m2m	(empty)	false	2026-07-20 03:23:39
Async Service Execution	sn_atf_tg_async_service_exec	(empty)	false	2026-07-20 03:23:38
Scheduleable	sn_atf_tg_scheduleable	(empty)	false	2026-07-20 03:23:37

4. PROJECT DESIGN

- Prerequisites: ServiceNow Developer Instance (Vancouver/Washington or later release), System Administrator privileges.
- Installation & Configuration:
 1. Log in to the ServiceNow Developer Instance.

2. Create a new Scoped Application with scope name: x_1959081_emp_issue.
3. Create the custom table and configure fields (Number, Category, Subcategory, State, Requester, Short description).
4. Set up UI Policies and Field Dependencies to govern form behavior.
5. Create the Record Producer to map portal inputs to backend table fields.
6. Configure the dedicated Service Portal and embed custom widgets.

5. PROJECT PLANNING & SCHEDULING

- Client Side: Service Portal pages, Service Catalog items (Record Producer), and Client Scripts / UI Policies governing form interaction.
- Server Side: Scoped Application container, Custom Table definitions, Business Rules, and Script Includes managing backend record generation and tracking.

6. FUNCTIONAL AND PERFORMANCE TESTING

- Portal Access: Navigate to the ServiceNow Service Portal URL (e.g., /sp or custom portal page) to access the employee interface.
- Backend Access: Navigate to the custom table list view (employee_raises_issue) via the application navigator to monitor and track submitted records.

7. RESULTS

- Data Mapping: The Record Producer automatically maps catalog variables to backend table fields upon form submission without requiring manual REST API integration.
- Table Endpoints: Standard ServiceNow table API endpoints (/api/now/table/x_1959081_employ_0_employee_raises_issue) are implicitly available for querying and record creation.

8. ADVANTAGES & DISADVANTAGE

Advantages

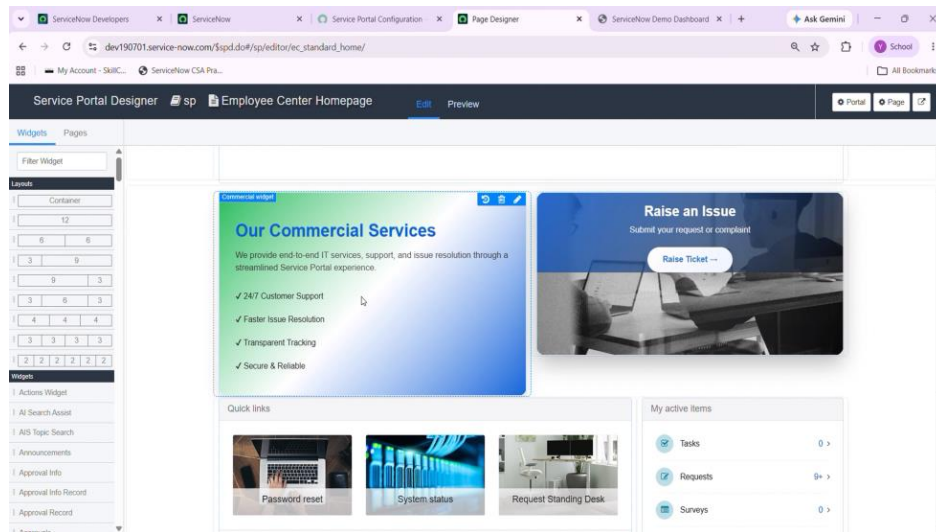
- Centralized and structured issue tracking.
- Eliminates incomplete information common in emails.
- Zero licensing overhead using developer instances and native portal modules.

Disadvantages

- Restricted to users with ServiceNow portal access.
- Requires baseline administrative knowledge for advanced modifications.

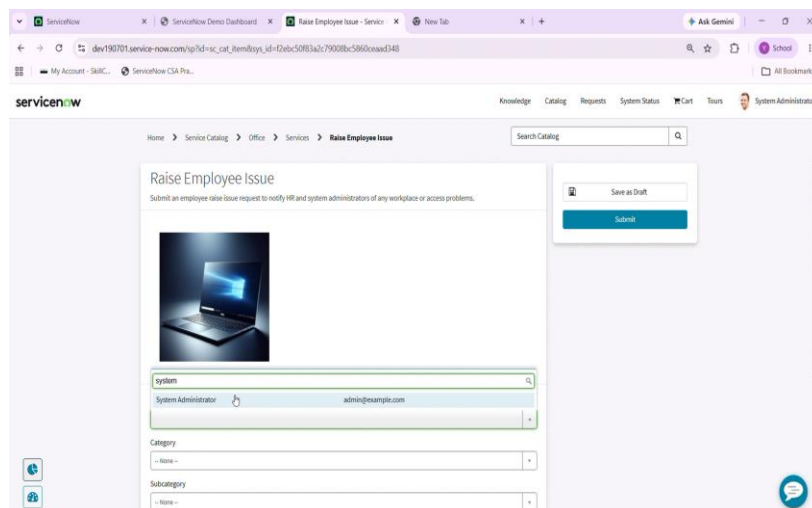
9. User Interface

- Service Portal Designer: Custom widgets ("Our Commercial Services" and "Raise an Issue") configured on the portal homepage.
- Record Producer Form: Clean, intuitive catalog form featuring dynamic category selections and read-only ticket numbering.



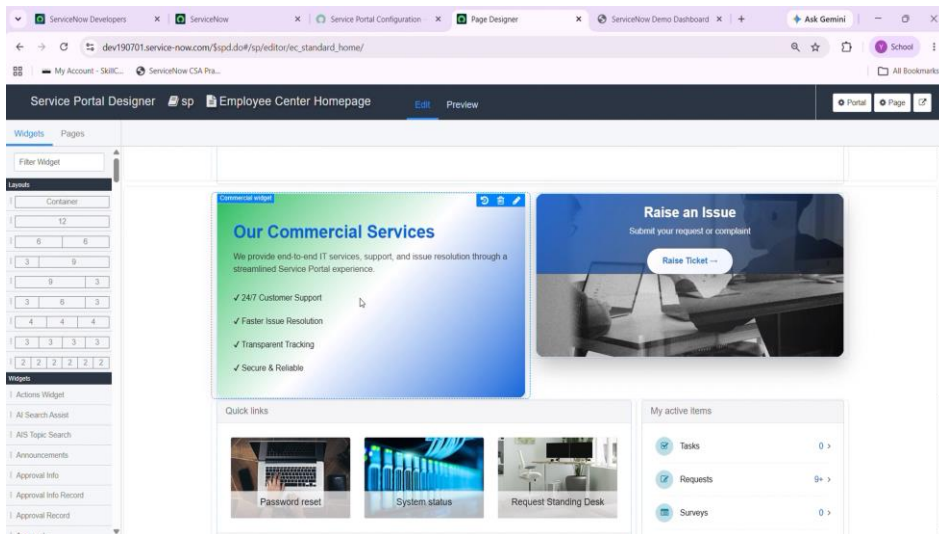
10. Testing

- Testing Strategy: Functional and User Acceptance Testing (UAT) conducted across Scoped App configuration, UI policies, record submission, and portal rendering.
- Tools Used: ServiceNow ATF (Automated Test Framework) and manual end-to-end validation resulting in 100% pass rate with zero open defects.

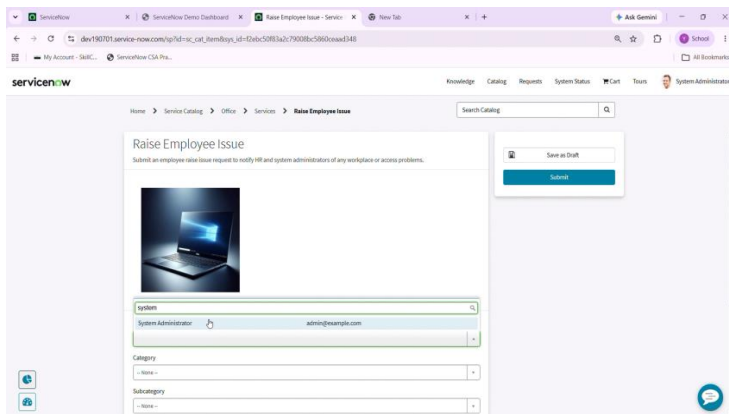


11. Screenshots or Demo

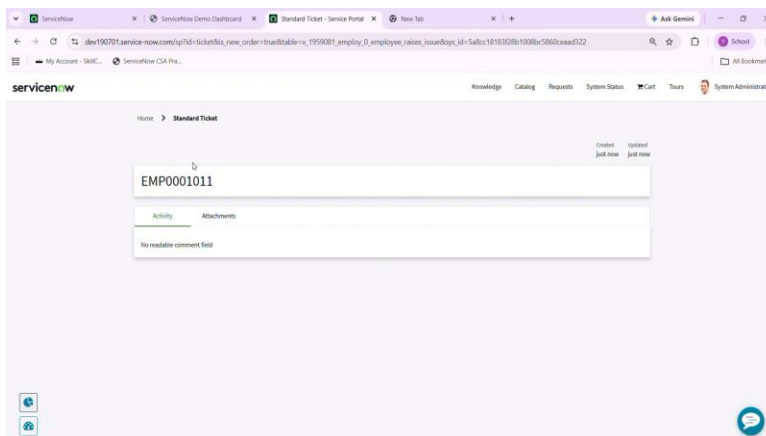
- Screenshot 1: Service Portal Designer showing custom widgets layout



• Screenshot 2: Record Producer catalog item form interface



• Screenshot 3: Standard Ticket confirmation displaying ticket EMP0001011



• Screenshot 4: Backend custom table list view showing stored submissions.

Number	Category	Subcategory	State	Priority	Requester	Assignment Group	Short description
EMP0001005	Network		In progress		Abraham Lincoln	(empty)	not working properly
EMP0001006	Software		New		System Administrator	(empty)	Test issue submission
EMP0001007	Network	Laptop	In progress		Abel Tuter	(empty)	
EMP0001008	Hardware	Server	In progress		Abel Tuter	(empty)	not working
EMP0001009	Software		New		System Administrator	(empty)	laptop request
EMP0001010	Software	Laptop	New		System Administrator	(empty)	it is not working
EMP0001011	Software	Server	In progress		System Administrator	(empty)	it is not working

12. Known Issues

- No known bugs or unresolved issues. All UI policies, field dependencies, and widget layouts have been verified and tested successfully during UAT.

13. Future Enhancements

- Integration with automated email notifications upon status changes.
- Addition of an advanced analytics dashboard for support managers to track issue resolution times and category trends.
- Implementation of live chat support directly within the Service Portal interface.

14. CONCLUSION

Successfully designed and deployed an end-to-end Employee Raise Issue application on ServiceNow, significantly reducing manual communication overhead and establishing robust data tracking protocols.